

GAROUA, Cameroon

WMO: 648600

Lat: 9.336N

Lon: 13.370E

Elev: 794

StdP: 14.28

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	62.2	63.9	23.7	18.4	95.4	27.5	22.0	93.2	13.6	89.7	12.0	90.5	1.4	0	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	20.8	107.3	69.6	105.6	69.9	103.9	70.1	80.8	91.9	80.1	91.2	79.5	90.7	7.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	151.2	85.5	77.3	146.4	84.9	76.9	144.3	84.6	45.1	92.0	44.4	91.6	43.7	91.4	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
17.1	14.5	12.8	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	48.8	83.8	1.2	2.0	47.9	85.3	47.2	86.4	46.6	87.6	45.7	89.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.5	79.4	85.3	92.0	93.4	88.9	83.9	81.7	80.6	81.0	84.5	84.4	79.3
	DBStd	5.67	3.90	4.54	3.19	3.45	4.31	3.81	3.04	2.74	2.75	2.46	2.68	3.67
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12599	913	989	1302	1302	1204	1016	984	948	931	1069	1032	909
	CDD65	7124	448	569	837	852	739	566	519	483	481	604	582	444
	CDH74	90315	5424	7952	13277	13290	10400	6651	5259	4324	4511	6847	7198	5182
Wind	WSAvg	5.2	3.5	3.8	5.3	7.3	7.0	6.7	6.3	5.7	5.0	4.7	3.7	3.5
	PrecAvg	39.9	0.0	0.0	0.1	1.8	4.4	5.5	7.7	9.7	7.9	2.8	0.1	0.0
	PrecMax	58.9	0.0	0.0	1.2	6.2	8.5	13.3	14.8	19.5	13.3	8.4	1.4	1.5
	PrecMin	18.4	0.0	0.0	0.0	0.0	0.2	1.1	2.9	3.5	2.1	0.1	0.0	0.0
	PrecStd	8.1	0.0	0.0	0.2	1.6	2.1	2.3	2.7	3.3	2.6	2.0	0.2	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	102.5	107.3	108.4	109.4	105.8	100.0	95.3	93.0	93.5	99.2	102.5	100.8
		MCWB	66.2	66.9	67.9	71.2	76.2	78.7	78.9	78.6	79.8	73.0	69.0	66.3
	2%	DB	100.0	104.3	107.3	107.4	102.7	97.0	92.9	91.1	91.8	97.1	100.4	98.4
		MCWB	65.1	66.4	68.3	71.2	77.2	78.8	78.6	78.1	79.1	74.8	69.0	65.9
	5%	DB	97.0	102.2	105.7	105.6	100.7	94.8	91.0	89.4	90.6	95.1	98.6	96.2
		MCWB	64.0	66.4	67.8	72.4	77.7	78.7	78.2	77.7	78.7	76.6	69.2	64.9
	10%	DB	93.6	99.1	104.0	103.7	98.5	92.0	89.0	87.6	88.9	93.1	96.9	93.2
		MCWB	62.6	64.9	67.3	73.6	77.5	78.3	77.5	77.3	78.1	78.1	69.4	63.7
	0.4%	WB	72.8	75.1	78.0	80.2	81.5	81.4	80.7	79.8	80.9	81.6	78.9	74.4
		MCDB	93.5	96.1	98.3	100.8	94.8	93.5	90.6	88.2	90.2	91.2	91.2	91.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	68.2	72.3	76.5	78.9	80.1	80.1	79.2	79.0	80.0	80.7	77.6	70.8
		MCDB	92.9	94.2	96.7	98.5	94.4	91.6	88.8	87.6	89.0	90.4	90.5	89.1
	5%	WB	65.0	68.8	75.3	78.0	79.5	79.4	78.7	78.3	79.2	79.9	76.2	66.6
		MCDB	94.4	96.2	95.3	96.9	93.8	90.6	88.1	86.6	87.8	89.6	88.5	91.0
	10%	WB	63.3	65.9	73.8	77.2	78.9	78.8	78.1	77.7	78.6	79.1	74.8	64.8
		MCDB	92.1	97.7	93.9	95.3	93.1	89.8	86.8	85.5	86.9	88.8	85.9	90.3
Mean Daily Temperature Range	5% DB	MDBR	29.7	28.3	24.9	20.8	18.2	15.9	14.0	13.3	14.8	18.2	27.0	28.9
		MCDBR	32.5	30.1	26.5	22.6	20.7	18.8	16.5	15.9	17.3	20.7	28.6	30.9
		MCWBR	10.7	10.5	9.6	8.0	5.9	6.1	5.6	5.4	6.7	6.8	8.0	9.2
	5% WB	MCDBR	31.6	27.0	21.3	20.3	18.5	17.2	15.7	14.8	16.5	17.6	22.2	28.9
		MCWBR	11.5	11.8	8.6	6.5	6.4	6.5	6.1	5.8	6.8	7.2	8.0	10.0
Clear-Sky Solar Irradiance	taub		0.556	0.681	0.747	0.700	0.595	0.556	0.508	0.491	0.508	0.523	0.480	0.488
	taud		1.818	1.570	1.488	1.633	1.861	1.987	2.150	2.210	2.139	2.070	2.091	1.995
	Ebn at Noon		241	217	205	213	232	238	251	259	258	253	261	257
	Edh at Noon		68	89	98	84	65	57	49	47	50	53	51	56
All-Sky Solar Radiation	RadAvg		1857	1906	1996	1961	1863	1725	1612	1509	1666	1862	1908	1834
	RadStd		53	66	102	89	90	67	79	75	69	54	49	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page